Advanced Computer Graphics (ACG)

Theory Session 8 – Neural Radiance Fields (NeRFs) and Presentation Topics

2026-2027

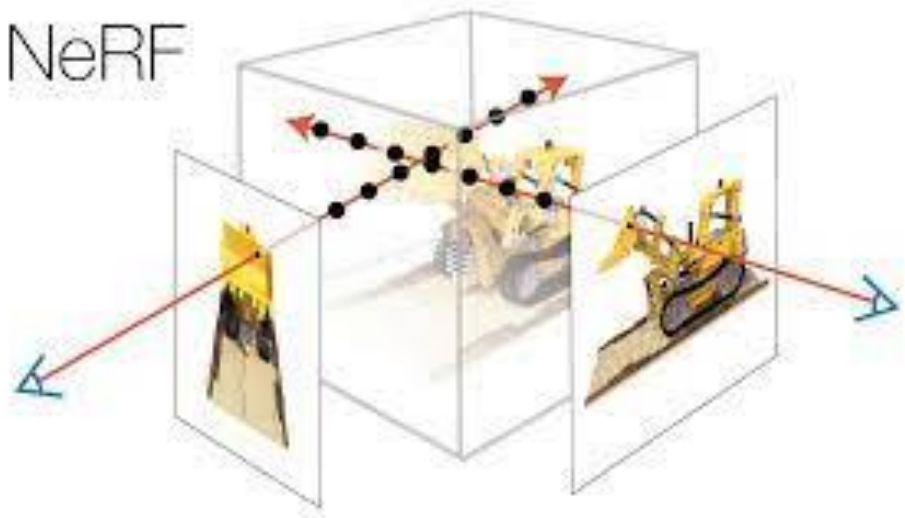
Pompeu Fabra University

Ricardo Marques (ricardo.marques@upf.edu)

Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Introduction & Motivation



NeRFs

Publication

"Representing Scenes as Neural Radiance Fields for View Synthesis"

(B. Mildenhall et al., ECCV 2020)

Impact

- Large number of citations
- Large number of follow-up papers
- Seminal paper which opened-up a new research path

Task: View Synthesis

Given one or more images of an object, create new (synthetic) views of that same object

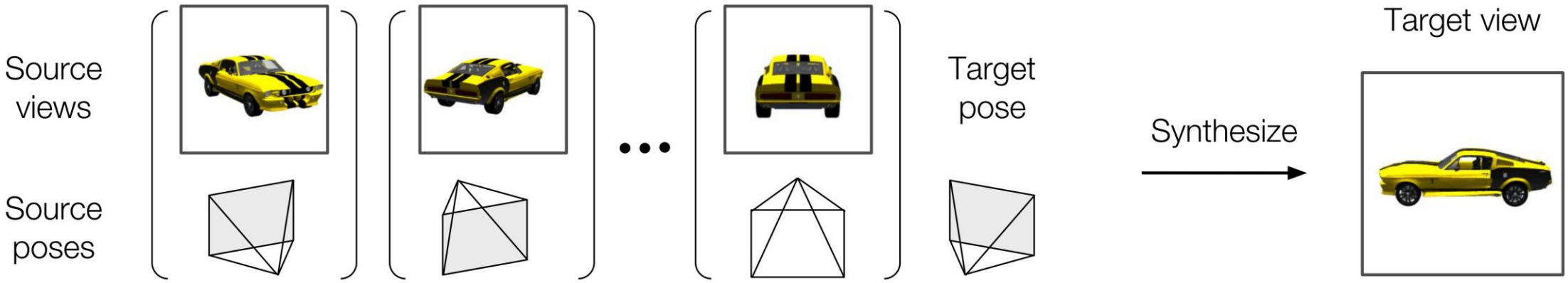


Image: [“Multi-view to Novel view: Synthesizing Novel Views with Self-Learned Confidence ”] **(Sun et al., ECCV 2018)**

View Synthesis using NeRFs

Problem Set-up

Consider a **pre-trained** NeRF model (fit to a particular scene)

- Predicts color and density at each 3D location, for each particular direction

Input:

x, y, z	3D scene location
θ, ϕ	Radiance flow direction

Output:

r, g, b	Color
σ	Density

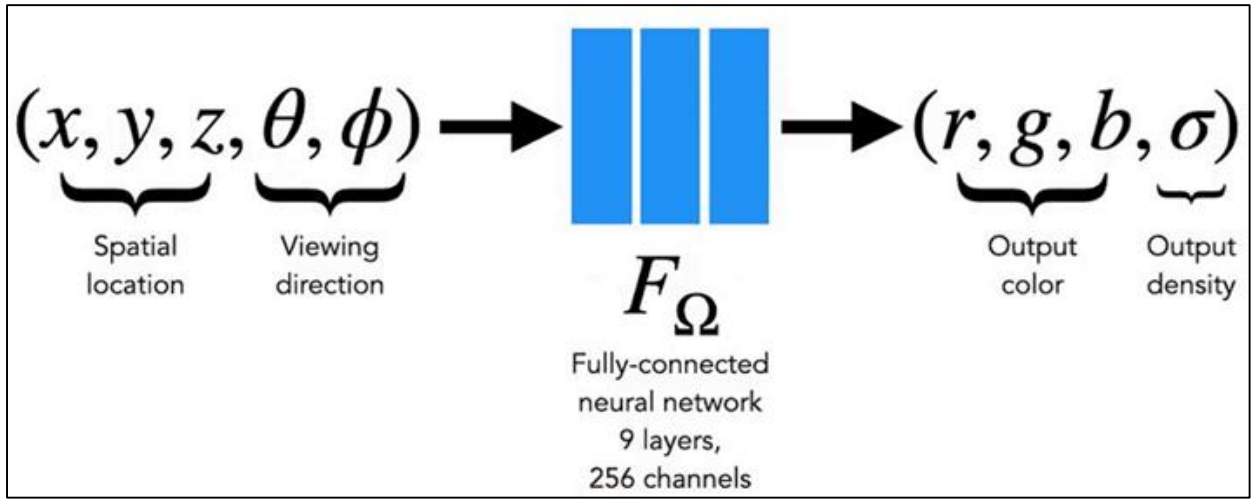
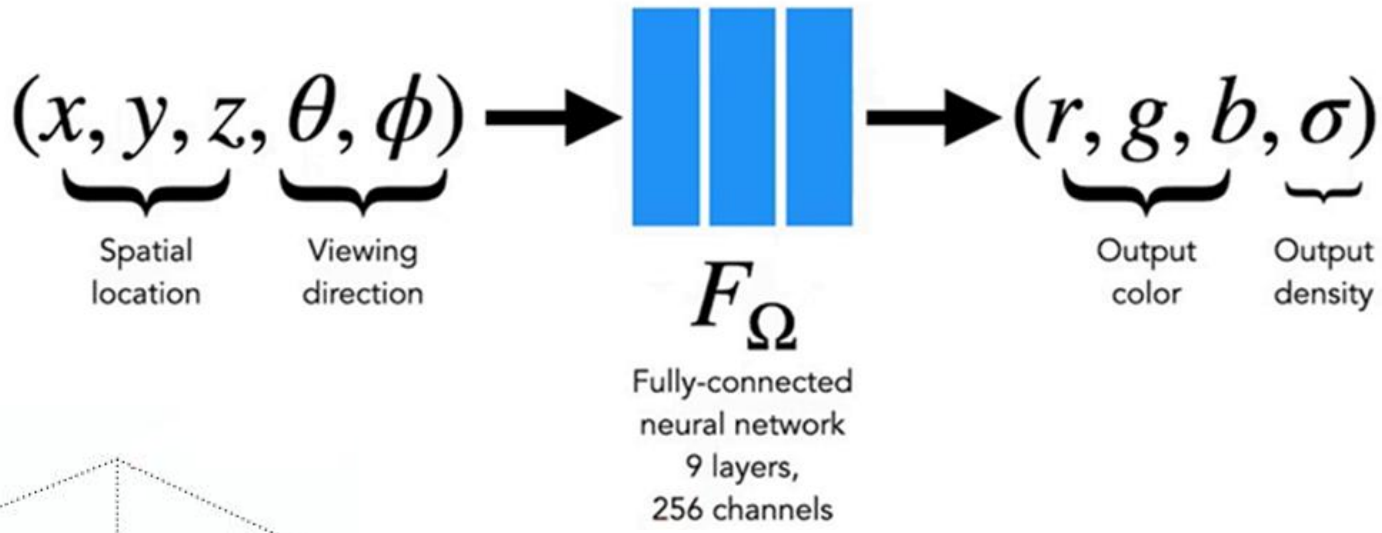
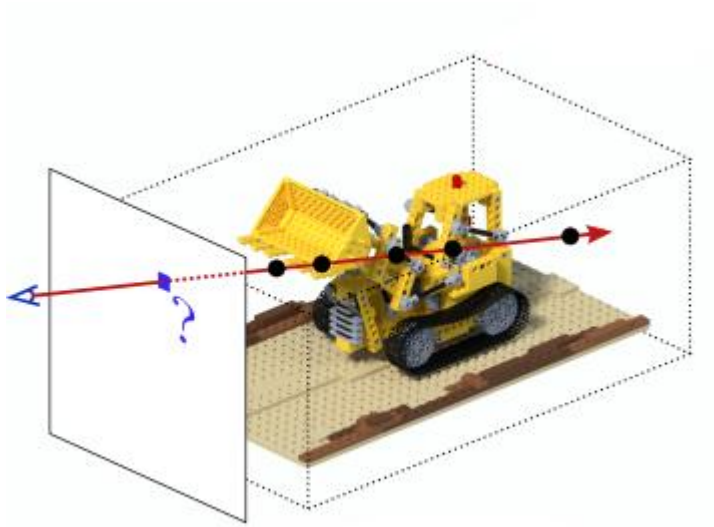
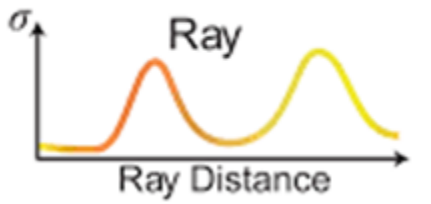


Image Synthesis using NeRFs



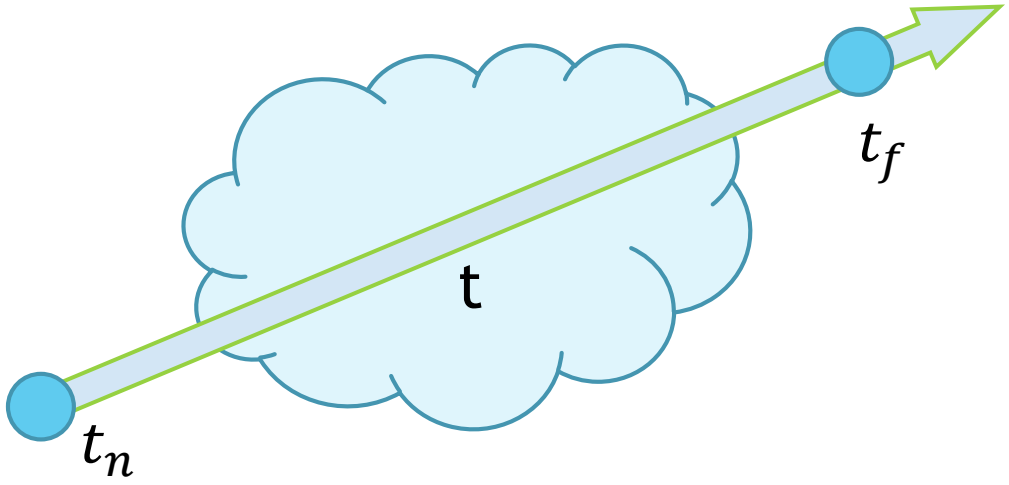




Volume Rendering and NeRFs (I)

Volume Rendering Equation (VRE)

$$C(r) = \int_{t_n}^{t_f} \sigma(r(t))\, c(r(t))\, T(t)\, \mathrm{d}t$$



where

$\sigma(r(t))$ is the absorption coefficient (scalar)

$c(r(t))$ is the emissive radiance (color)

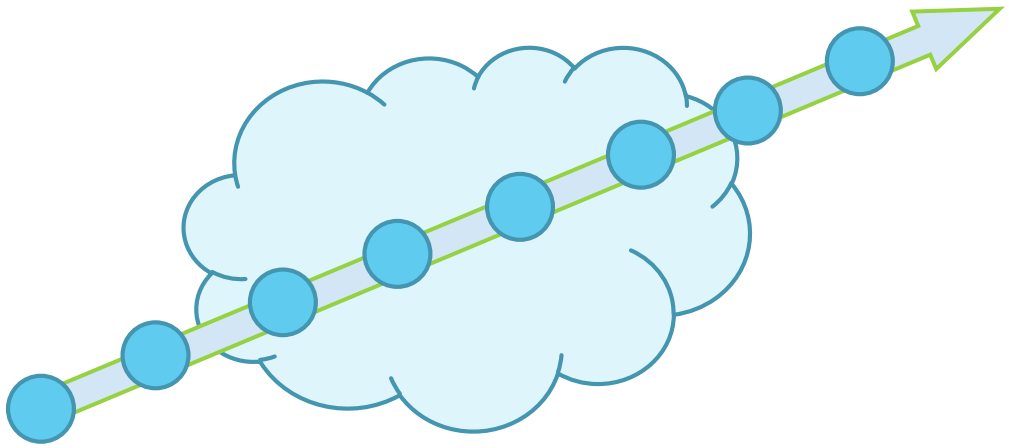
$\mathrm{T(t)} = \exp\left(-\int_{t_n}^{t} \sigma(r(s))\, \mathrm{d}s\right)$ is the transmittance

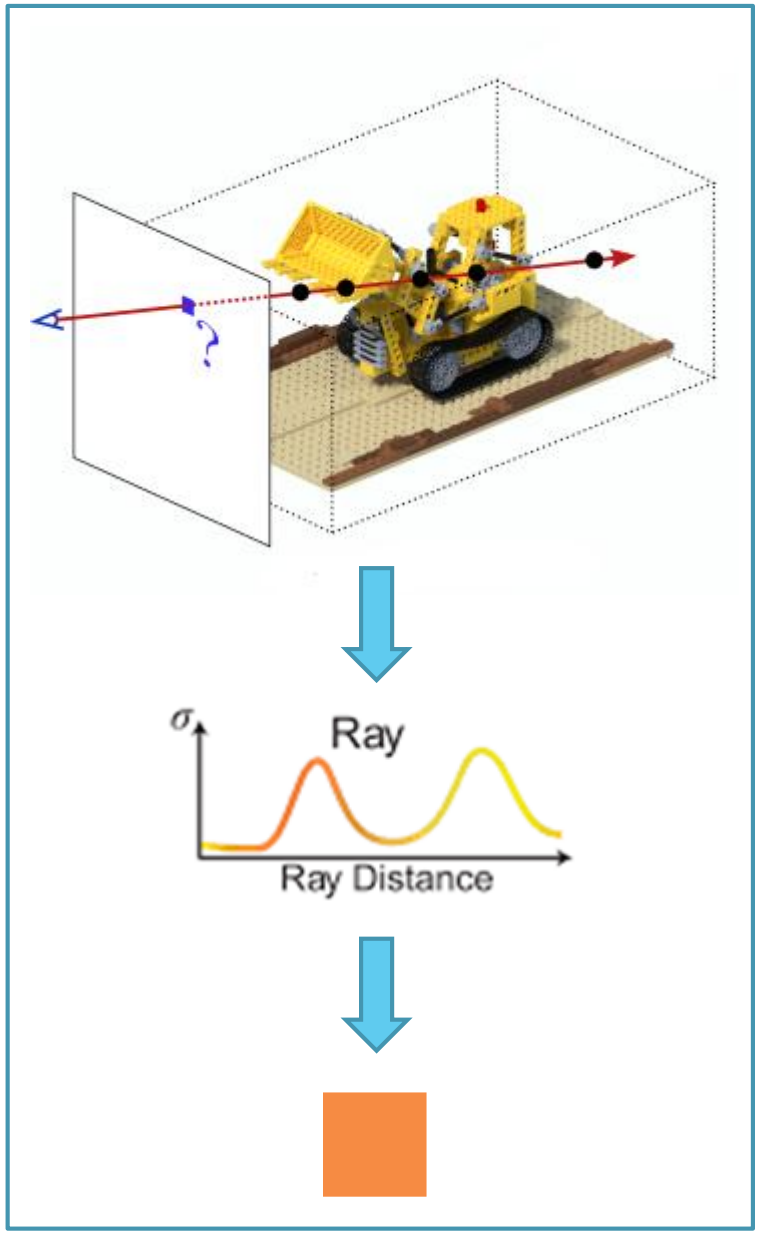
Volume Rendering and NeRFs (II)

$$C(r) = \int_{t_n}^{t_f} \sigma\big(r(t)\big)\, c\big(r(t)\big)\, T(t)\, \mathrm{d}t$$

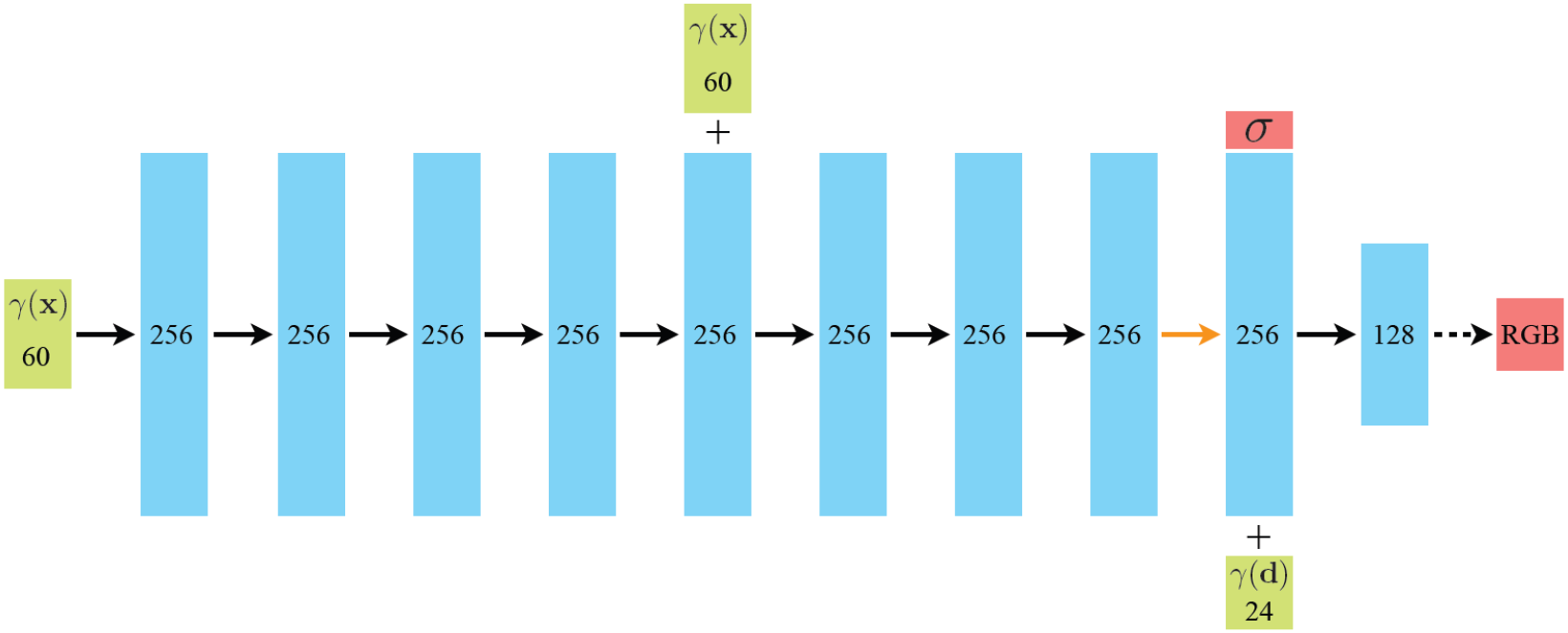
Discretize the VRE to compute its approximate value:

$$C(r) \approx \hat{C}(r) = \sum_{i=1}^{N} (1 - \exp(-\sigma_i\, \delta_i))\, c_i\, T_i$$



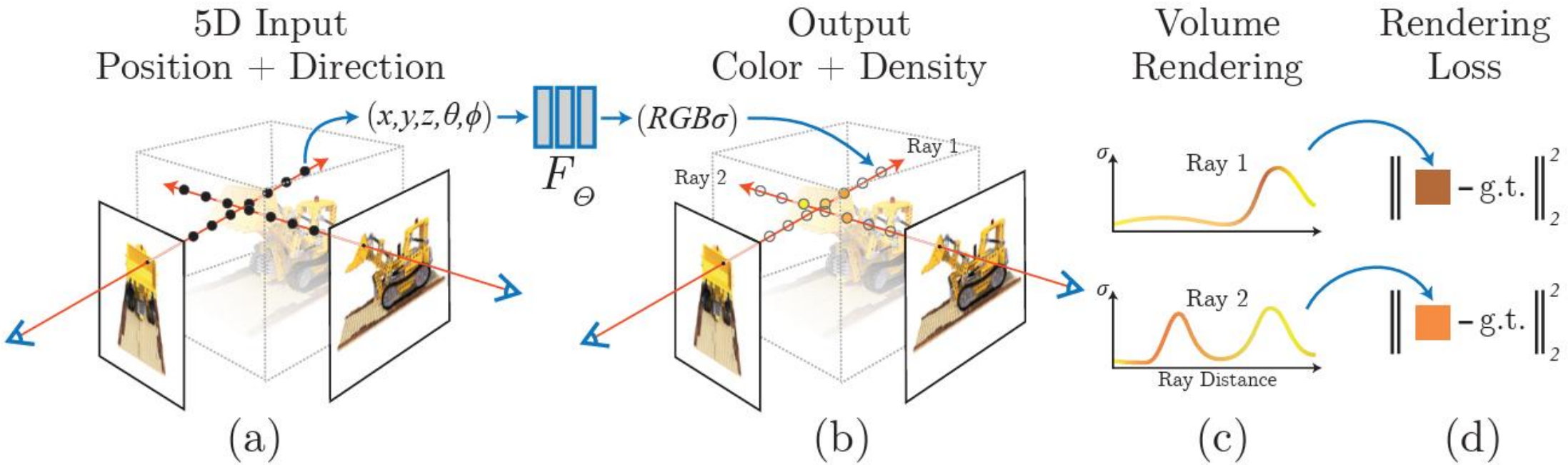


Model and Training



Neural Network Architecture

Training


5D Input
Position + Direction
(x,y,z,θ,ϕ)
F_Θ
Output
Color + Density
$(RGB\sigma)$
Ray 1
Ray 2
Volume
Rendering
Rendering
Loss
σ
Ray 1
Ray 2
Ray Distance
$\| - \text{g.t.} \|_2^2$
(a)
(b)
(c)
(d)

Training Procedure Overview

Training is performed over batches rays (can be from different images)

Use hierarchical sampling to query coarse and fine points

Apply the VRE to compute the pixel color

Compute the L_2 norm between the computed color and the reference

Optimize the neural network parameters

Implementation Details

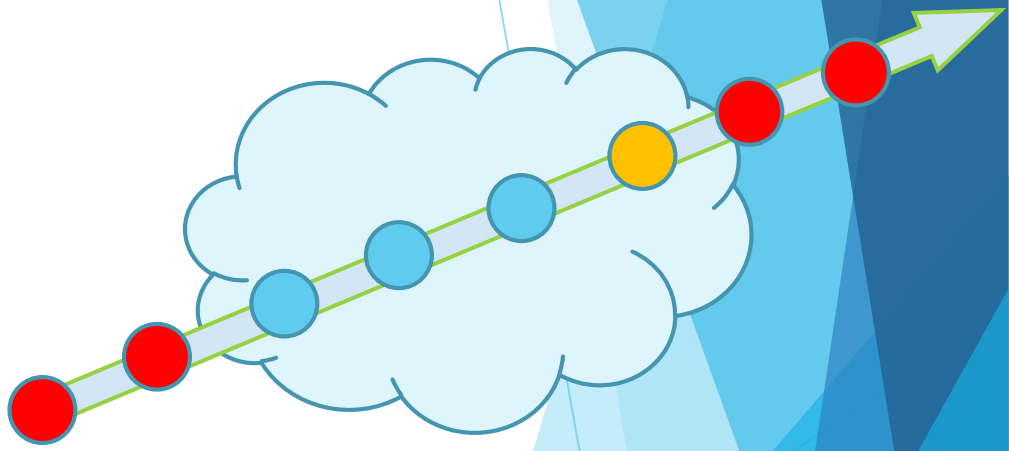
Hierarchical Sampling for Improved Performance

Observation: uninformed uniform sampling might be inefficient

- Outside relevant parts of the volume (red)
- Occluded (orange)

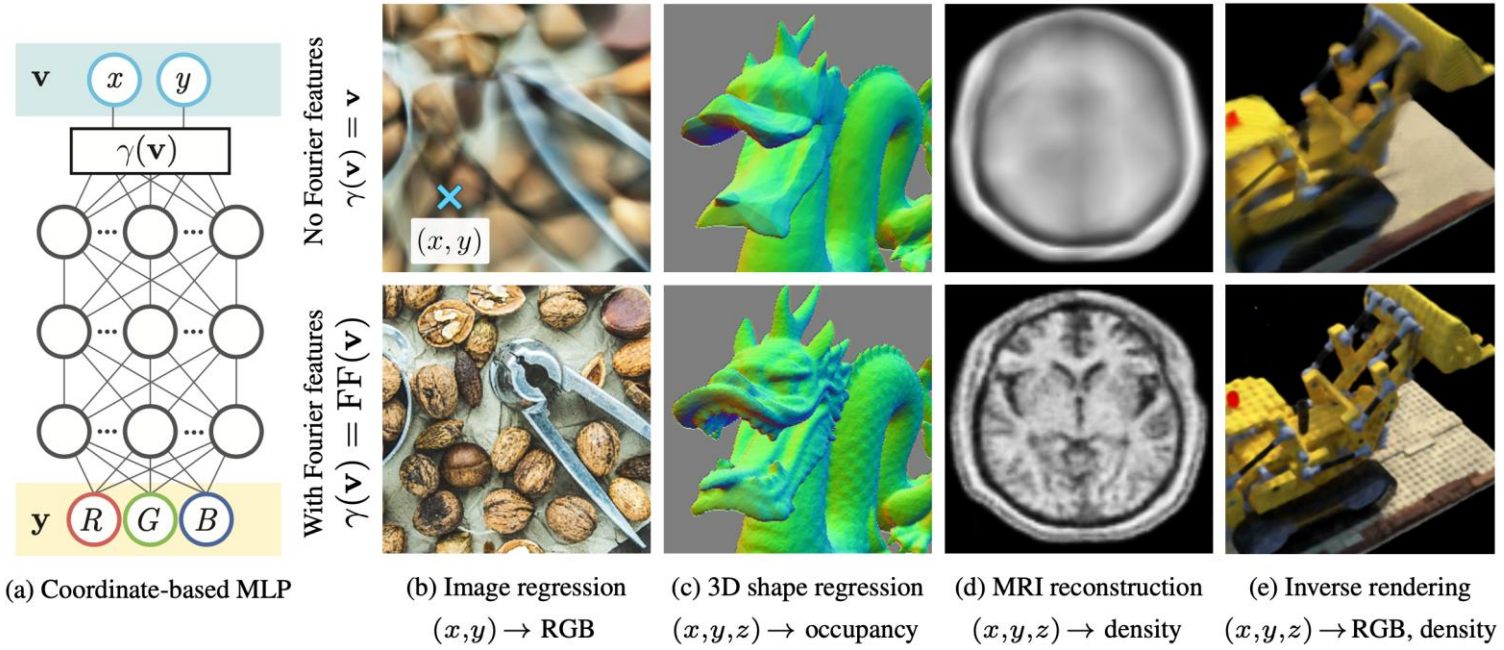
Strategy: use a coarse and a fine network

- Uniform sampling of the coarse network to obtain a rough estimate (low frequency)
- Further refine the sampling in relevant parts of the volume



Positional Encoding

Map individual components of position and direction vectors to a higher dimensional space



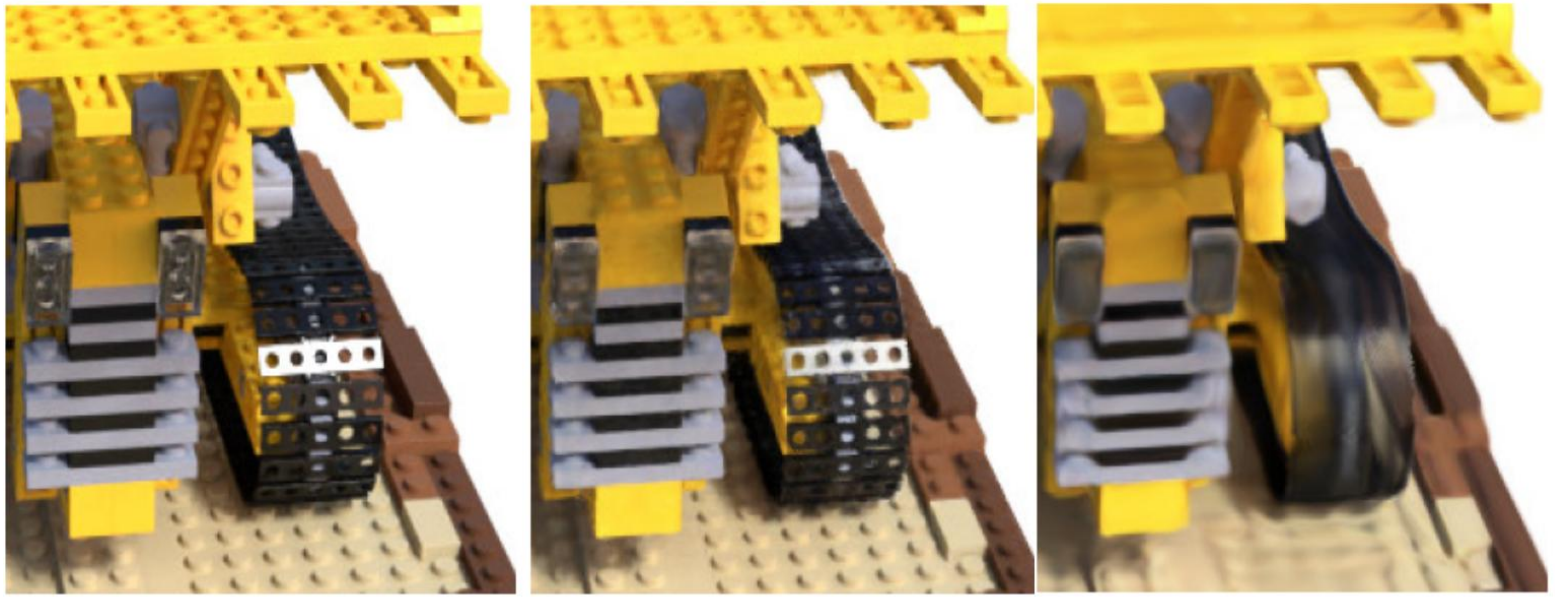
(a) Coordinate-based MLP

(b) Image regression $(x,y) \rightarrow$ RGB

(c) 3D shape regression $(x,y,z) \rightarrow$ occupancy

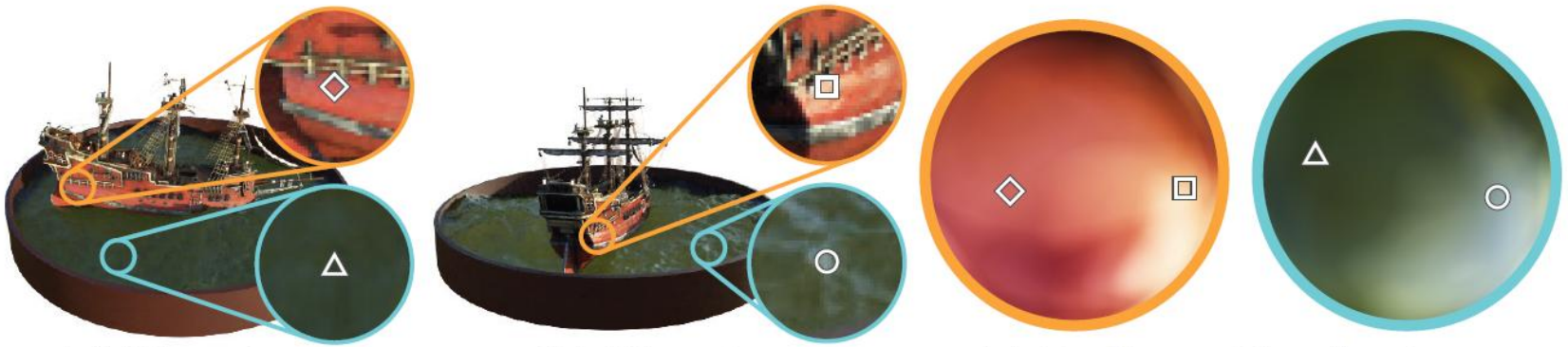
(d) MRI reconstruction $(x,y,z) \rightarrow$ density

(e) Inverse rendering $(x,y,z) \rightarrow$ RGB, density



Positional Encoding Results

Visual Results



(a) View 1 (b) View 2 (c) Radiance Distributions

Results: View-dependent radiance



Results: High Frequency Details



Results: 360 degrees



Results: Depth Extraction

Conclusions

Wrapping Up

We have seen an overview of the original NeRF approach

- Problem set-up tailored to volume rendering approach
- Sampling strategy
- Importance of positional encoding
- Training approach and neural network architecture

Main advantages:

- Small memory footprint compared to previous approaches

Main limitations:

- Lack of generalization
- Data hungry: 100 to 500 images per scene required (further works tackle this limitation)

A Sample of Follow-up Papers

Enhanced positional encoding:

"Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains" (Tancik et al., NeurIPS 2020)

Multi-scale representations:

"Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields" (Barron et al., ICCV 2021)

Learned Initializations:

"Learned Initializations for Optimizing Coordinate-Based Neural Representations" (Tancik et al., CVPR 2021)

Bibliography

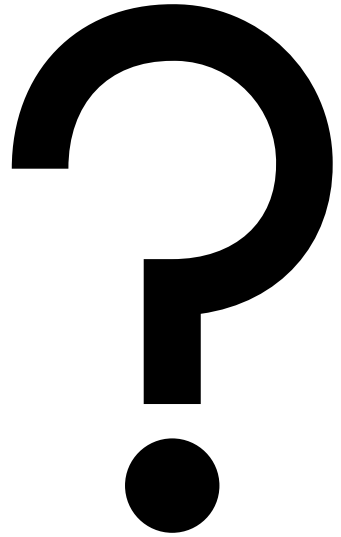
Paper: https://arxiv.org/abs/2003.08934

Project page: https://www.matthewtancik.com/nerf

Tutorial EVVC 2020:
https://sites.google.com/berkeley.edu/nerf-tutorial/home

Presentations:

- Elvin Yang, CS391R: Robot Learning (Fall 2022), University of Texas
- David Lindell, CSC2529: Radiance Fields Representations, University of Toronto



Questions

Topics for Final Project and Presentation

Overall Goal

- Group work around a CG topic chosen by you:
  - Understand and present the topic and its background
  - Introduce useful tools to work on it
  - Eventually add a prototype/proof-of-concept implementation in any of the two projects delivered (CPU or GPU, highly appreciated)

Choosing the Topic

- Each group should choose a topic as soon as possible
  - You can choose your own topic (of advanced computer graphics)
  - Or one of the suggested topics listed afterwards
  - Each topic can only be presented by a single group
  - The first group to ask for a topic, will be the one presenting it (first come, first served)

- Task: send an email (with everyone in the group in copy) with:
  - The **group number**
  - The **topic** chosen and **why** it has been chosen

- **Deadline for Choosing the Topic:** 5pm Nov 10

List of Suggested Topics (Order is Irrelevant)

1. Photon Mapping
2. Irradiance Caching and Radiance Caching
3. Virtual Point Lights (VPLs) for Global Illumination
4. Volumetric Path Tracing
5. Multiple Importance Sampling for BRDFs and Area Light Sources
6. Path Guiding
7. Ambient Occlusion
8. Motion Blur and Depth of Field
9. Tone Mapping
10. Microfacet BRDFs
11. Subsurface Scattering
12. Acceleration Structures for Ray Tracing
13. Efficient Rendering of Dynamic Scenes
14. Denoising Path-Traced Images
15. Fresnel Reflection
16. Volume Rendering for Medical Data

1st Presentation (Outline)

- The format suggested is 3-5 slides
  - Motivation and relevance of the topic
  - Application examples
  - Brief sketch of the topic (key idea, issues at stake)
  - Scheme of intended contents
  - References
- Each Group
  - Submit slides by 10am of the presentation day (November 24) – Task Aula Global
  - Present the outline: 5 minutes + 3 minutes of questions
- Each person will provide a concise feedback to each outline
  - Task in Aula Global for that

2nd Presentation (The Actual Topic Content)

- The slides should contain
  - Relevance
  - Issues addressed
  - Key points to fix the issue with enough detail to understand the solution
  - A specific point explained in detail to understand it
  - Resources: code resources, documentation resources, other
  - Visual material, preferably demo
  - Slide notes (or a *google doc* should exist), so that the material is self-contained
  - Use as many slides as needed

- **IMPORTANT**: a prototype/proof-of-concept implementation in one of the course's frameworks will be highly appreciated

2nd Presentation (The Actual Topic Content)

- Each Group
  - Submit slides by 10am of the FIRST presentation day (December 3) – Task Aula Global
  - SAME DEADLINE FOR ALL GROUPS
  - Present the topic: total time to be defined, equal distribution of time among group members
- Each person will provide a concise feedback to each presentation
  - Task in Aula Global for that

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The End

Questions?